

MCKV Institute of Engineering
243 G.T. Road(N), Liluah
Howrah-711204

Computer Organisation Lab.
CS 393

Experiment No. : 1

Title : **Familiarity with the state of IC chips of**
a. Multiplexer, b. Decoder, c. Encoder, d. Comparator

Theory:

a) Multiplexer: Multiplexer is a combinational ckt. that is one of the most widely used in digital design. The multiplexer is a data selector which gates one out of several i/ps to a single o/p. It has n data i/ps & one o/p line & m select lines where $2^m = n$. Depending upon the digital code applied at the select inputs one out of n data input is selected & transmitted to a single o/p channel. Normally strobe(G) input is incorporated which is generally active low which enables the multiplexer when it is LOW. Strobe i/p helps in cascading. No. of ICs are available such as 74157, 74158 (Quad 2:1 mux), 74352, 74153 (dual 4:1 Mux.), 74151A, 74152 (8:1 Mux.), 74150 (16:1 Mux). IC 74151A is a 8 : 1 multiplexer which provides two complementary o/ps Y & Y'. The o/p Y is same as the selected i/p & Y' is its complement. The n:1 multiplexer can be used to realize a m variable function. ($2^m = n$, m is no. of select inputs).

b) We shall deal with BCD – to Decimal Decoder: A decoder is similar to a demultiplexer, with one exception- there is no data input. The only inputs are the control bits DCBA. BCD is abbreviation for binary-coded decimal. The BCD code expresses each digit in a decimal number by its nibble equivalent. Note that BCD digits are from 0000 to 1001

Encoder: An encoder converts an active input signal into a coded output signal. 74147 is a decimal – to –BCD priority encoder. It is called Priority encoder because it gives priority to the highest-order input.

Comparator: Magnitude Comparator compares two binary data signals A & B and generates the results of comparison in form of three output signals $A > B$, $A = B$, $A < B$. IC 7485 is a 4-bit comparator. The cascade inputs $A > B$, $A = B$, $A < B$ can be used to construct a comparator comparing more than 4 bits. The compare outputs depend on both compare i/p s as well as cascade i/p s.

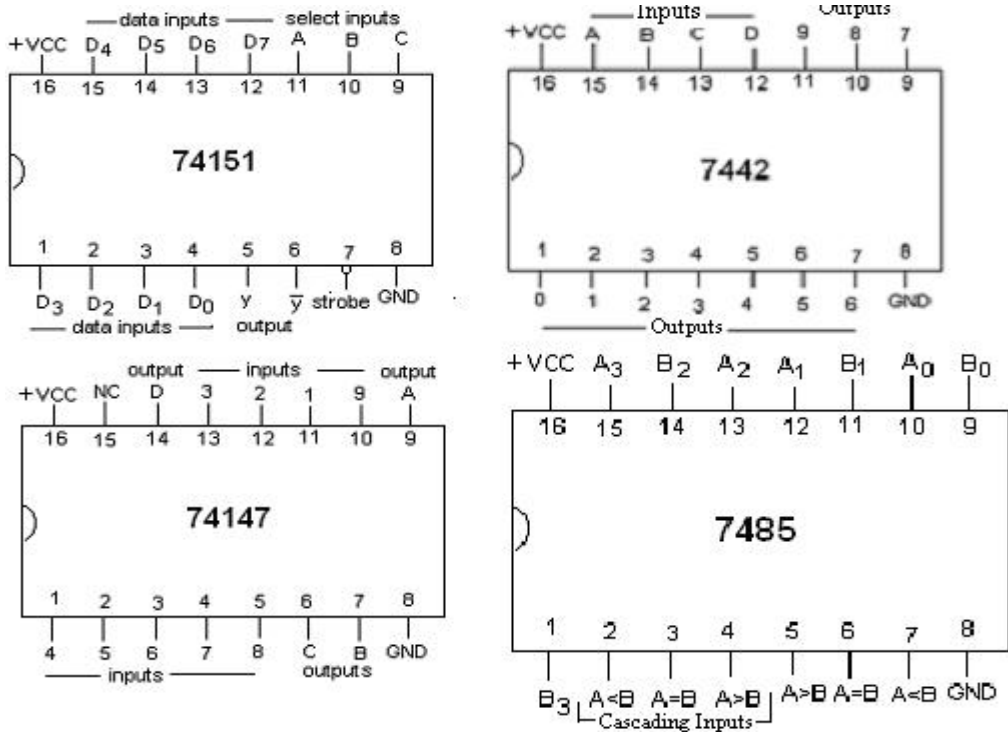
Equipments & Components Required :

1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components
 - a. ICs 74151, 7442, 74147 & 7485.
 - b. Resistors.
 - c. LEDs

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Pin Diagram of IC -74151, 7442, 74147 & 7485.



Experimental Procedure:

- 1) Give biasing to the IC.
- 2) Do necessary connections.
 (Low levels refer to 0 volts and high levels refer to +5volts)
- 3) Verify the truth table for respective IC's for its operation

Observation Table for 8:1 MUX

Strobe	Input			output	
	Select Inputs			Y	Y'
ST	C	B	A		
0	0	0	0		
0	0	0	1		
0	0	1	0		
0	0	1	1		
0	1	0	0		
0	1	0	1		
0	1	1	0		
0	1	1	1		
1	X	X	X		

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Observation Table of BCD-to-Decimal Decoder

Sl No.	BCD Inputs				Low Active Decimal Outputs									
	D	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9
0.	0	0	0	0										
1.	0	0	0	1										
2.	0	0	1	0										
3.	0	0	1	1										
4.	0	1	0	0										
5.	0	1	0	1										
6.	0	1	1	0										
7.	0	1	1	1										
8.	1	0	0	0										
9.	1	0	0	1										

Observation Table of Encoder

Active low decimal inputs									Active low BCD outputs			
1	2	3	4	5	6	7	8	9	D	C	B	A
1	1	1	1	1	1	1	1	1				
0	1	1	1	1	1	1	1	1				
X	0	1	1	1	1	1	1	1				
X	X	0	1	1	1	1	1	1				
X	X	X	0	1	1	1	1	1				
X	X	X	X	0	1	1	1	1				
X	X	X	X	X	0	1	1	1				
X	X	X	X	X	X	0	1	1				
X	X	X	X	X	X	X	0	1				
X	X	X	X	X	X	X	X	0				
X	X	X	X	X	X	X	X	X				

Observation Table for 4-bit Comparator

A ₃	A ₂	A ₁	A ₀	B ₃	B ₂	B ₁	B ₀	A>B	A=B	A<B

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Experiment No. : 2

Title : Design an Adder/Subtractor composite unit.

Equipments & Components Required:

1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components
 - a. ICs 7483,7486.
 - b. Resistors.
 - c. LEDs.

Theory:

a) Addition:

IC type 7483 is a 4-bit binary adder with carry. The pin assignment is shown in Fig 1. The two 4-bit input binary numbers are A_0 through A_3 and B_0 through B_3 . The 4-bit sum is obtained from S_0 through S_3 . C_i is the input carry and C_o is the out carry. This IC can be used as an adder-subtractor.

b) Subtraction:

The subtraction of two binary numbers can be done by taking the 2's complement of the subtrahend and adding it to the minuend. The 2's complement can be obtained by taking the 1's complement and adding 1. To perform $A - B$, we complement the four bits of B, add them to the four bits of A, and add 1 to the input carry. Four XOR gates complement the bits of B when the mode select $M = 1$ (because $x \cdot 1 = x'$) and leave the bits of B unchanged when $M = 0$ (because $x \cdot 0 = x$) thus, when the mode select M is equal to 1, the input carry C_i is equal to 1 and the sum output is A plus the 2's complement of B. When M is equal to 0, the input carry is equal to 0 and the sum generates $A + B$.

Experimental Procedure:

- 1.Design the circuit of 4-bit adder –subtractor using 7483 and 7486.
2. Implement the circuit as per design.
3. Give biasing to the IC .
4. Perform the following operations and record the values of the output sum and the output carry C_o .

Observations:

Decimal A B	C_i	A_3	A_2	A_1	A_0	B_3	B_2	B_1	B_0	C_o	S_3	S_2	S_1	S_0
9+5														
9-5														
10+6														
6-10														
10-6														

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Experiment No. : 3

Title : Design a BCD adder.

Equipments & Components Required:

1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components--
 - a. ICs 7400,7410,7483.
 - b. Resistors.
 - c. LEDs.

Theory: When two BCD digits are added together with a possible carry from the previous stage, the sum can be maximum 19. Since, maximum BCD digit is 9, thus $9+9+1$ (carry from previous stage)=19. When two BCD digits are added using 4-bit binary adder, the sum will be in binary form and the range of sum will be 0 to 19 in equivalent decimal. The Binary Sum and the BCD sum in this range is shown in Table below. By observing the table it is clear that the two sums are identical upto decimal 9 and thus no correction is required. When sum is greater than 9, we find an invalid BCD sum representation. The binary 0110 (i.e. 6) is added to the binary sum to get the valid BCD sum representation. This will generate an output carry as required.

The BCD adder must be able to perform the following:

- (a) Add two BCD digits using one 4-bit parallel adder.
- (b) If the binary sum of this adder is greater than or equal to binary 1010 (i.e. decimal 10), add binary 0110 (decimal 6) to the sum and generates the required output carry.

From the table the logic circuit for necessary correction can be derived. It is clear that when binary sum from 4-bit parallel adder is greater than 01001(i.e. decimal9) we need a correction. This can be said in other word as BCD carry X will be in high state when either of the following conditions occurs:

1. When $C=1$ (sums greater than or equal to 16)
2. When $S_3=1$ and either S_2 or S_1 (Sums in the range 10 to 15). Since decimal numbers 8 and 9 also have $S_3=1$. To distinguish decimal numbers 10 to 15 from 8 and 9, we use $S_2 S_1$ bit positions.

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Relation of Binary Sum and BCD sum

Binary Sum					BCD Sum						Decimal
C	S ₃	S ₂	S ₁	S ₀		X	S ₃	S ₂	S ₁	S ₀	
0	0	0	0	0		0	0	0	0	0	0
0	0	0	0	1		0	0	0	0	1	1
0	0	0	1	0		0	0	0	1	0	2
0	0	0	1	1		0	0	0	1	1	3
0	0	1	0	0		0	0	1	0	0	4
0	0	1	0	1		0	0	1	0	1	5
0	0	1	1	0		0	0	1	1	0	6
0	0	1	1	1		0	0	1	1	1	7
0	1	0	0	0		0	1	0	0	0	8
0	1	0	0	1		0	1	0	0	1	9
0	1	0	1	0		1	0	0	0	0	10
0	1	0	1	1		1	0	0	0	1	11
0	1	1	0	0		1	0	0	1	0	12
0	1	1	0	1		1	0	0	1	1	13
0	1	1	1	0		1	0	1	0	0	14
0	1	1	1	1		1	0	1	0	1	15
1	0	0	0	0		1	0	1	1	0	16
1	0	0	0	1		1	0	1	1	1	17
1	0	0	1	0		1	1	0	0	0	18
1	0	0	1	1		1	1	0	0	1	19

Thus the condition for correction and the output carry can be expressed as:

$$X = C + S_3S_2 + S_3S_1$$

Experimental Procedure:

1. Design the circuit of BCD adder using 7483 and NAND gates.
2. Implement the circuit as per design.
3. Give biasing to the IC .
4. Perform the following operations and record the values of the output sum and output carry(X= Ten's) where A and B are two BCD numbers.

Observations:

A ₃	A ₂	A ₁	A ₀	B ₃	B ₂	B ₁	B ₀	X	S ₃	S ₂	S ₁	S ₀

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Experiment No. : 4

Title : Design a Carry-look ahead adder.

Equipments/ Components Required:

1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components
 - a. ICs 7408,7432,7486.
 - b. Resistors.
 - c. LEDs.

Theory: A Carry Look-ahead Adder (CLA) is a high-speed adder which adds two numbers without waiting for the carries from the previous stages. In the CLA, carry- inputs of all stages are generated simultaneously, without using carries from the previous stages. In the full adder, the carry output C_{i+1} is related to its carry input C_i as follows:

$$C_{i+1} = G_i + P_i C_i$$

Where $G_i = A_i B_i$ and $P_i = A_i + B_i$

The function G_i is called the carry

Experimental Procedure:

(The students will have to design 2 bit carry look ahead adder circuit)

1. Low levels refer to 0 volts and high levels refer to +5volts.
2. Implement the circuit as per design.
3. Verify the truth table.

Circuit Diagram:

Truth Table:

Observations:

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Experiment No. : 6
Objective : Use ALU chip for multibit arithmetic operation.

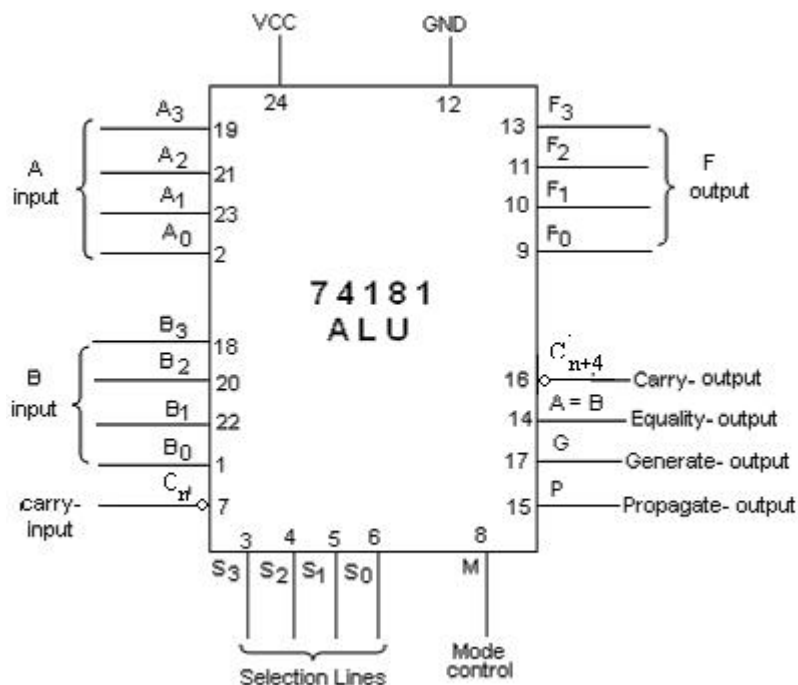
Equipments Required:

1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components
 - a. IC- 74181.
 - b. Resistors.
 - c. LEDs.

Experimental Procedure:

- 1.Low levels refer to 0 volts and high levels refer to +5volts.
2. Connect VCC to and Ground to ALU.
- 3.Verify the truth table.

Circuit Diagram:



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The block diagram of 74181 ALU is shown in Fig. This circuit can be used for both positive as well as negative logic. The functions of various input, output and control lines are given below.

Inputs

A and B : 4-bit binary data-input.

C_n : Carry-input. This is active-low, i.e., if the carry-input is to be given, it should be at logic '0'.

Outputs

F : 4-bit binary data-output.

C_{n+4}: Carry-output. For the addition-operation, a logic '0' on this line indicates a carry-output. For the subtraction-operation, it indicates the sign of the output. Logic '0' on this line indicates a positive result, and logic '1' indicates a negative result expressed in the 2's complement form.

A=B :Equality-output. Logic '1' on this line indicates A=B
(see also the comparator operation).

G : Carry-generate-output.

P : Carry-propagate-output.

Sl. No	Selection Line				Active high data		
					M=1; Logic functions	M=0; Arithmetic operations	
	S ₃	S ₂	S ₁	S ₀	C _n =X	C _n =1 (no carry)	C _n =0 (with carry)
0	0	0	0	0	F=A'	F=A	F= A PLUS 1
1	0	0	0	1	F=(A+B)'	F=A+B	F=(A+B) PLUS 1
2	0	0	1	0	F=A'B	F=A+B'	F=(A+B' PLUS 1)
3	0	0	1	1	F=0	F=MINUS 1 (2's COMPL)	F= ZERO
4	0	1	0	0	F=(AB)'	F=A PLUS AB'	F=A PLUS AB' PLUS 1
5	0	1	0	1	F=B'	F=(A+B) PLUS AB'	F=(A+B) PLUS AB' PLUS 1
6	0	1	1	0	F=A XOR B	F=A MINUS B MINUS 1	F=A MINUS B
7	0	1	1	1	F=AB'	F=AB' MINUS 1	F=AB'
8	1	0	0	0	F=A'+B	F=A PLUS AB	F=A PLUS AB PLUS 1
9	1	0	0	1	F=A XNOR B	F=A PLUS B	F=A PLUS B PLUS 1
10	1	0	1	0	F=B	F=(A+B') PLUS AB	F=(A+B') PLUS AB PLUS 1
11	1	0	1	1	F=AB	F=AB MINUS 1	F=AB
12	1	1	0	0	F=1	F=A PLUS A	F=A PLUS A PLUS 1

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13	1	1	0	1	$F=A+B'$	$F=(A+B)$ PLUS A	$F=(A+B)$ PLUS A PLUS 1
14	1	1	1	0	$F=A+B$	$F=(A+B')$ PLUS A	$F=(A+B')$ PLUS A PLUS 1
15	1	1	1	1	$F=A$	$F=A$ MINUS 1	$F=A$

G and P outputs are used when a number of 74181 circuits are used in cascade, alongwith the 74182 look-ahead carry-generator circuit to make the arithmetic operations faster.

Control lines

Selection Lines (S) : These are four function-select lines, to select any one of the operations.

Mode Control (M): This input determines whether the arithmetic (M=0) or logic (M=1) operation is performed

EXPERIMENTS

Logical Operations

Set up the circuit of the 74181 IC for logical operations (M=1). For each of the 16 lines of Table, apply the signals at the selection-lines (one-by-one) and verify the logical operations performed on two binary-data-inputs A and B.

Arithmetic Operations

Set up the circuit of the 74181 IC for arithmetic operations (M=0). Perform the following operations.

(i) Addition

Refer to line 9 of the Table. Apply 1001 signals at the selection-lines. For data-inputs A and B, observe the outputs at S_{n+4} , F_3 , F_2 , F_1 , F_0 for (a) $C_n=1$ (no carry) and (b) $C_n=0$ (with carry). The result will be a 5-bit binary word expressed as $(C_{n+4})F_3F_2F_1F_0$.

Try for various combinations of A and B. Verify the results.

(ii) Subtraction

Refer to line 6 of table. Apply 0110 at the selection-lines. For data-inputs A and B, observe the outputs at S_{n+4} , F_3 , F_2 , F_1 and F_0 for (a) $C_n=0$ (with carry) and (b) $C_n=1$ (no carry) . The result is expressed in sign-magnitude form. The C_{n+4} output gives the sign; it is 0 for positive result and 1 for negative result. $F_3F_2F_1F_0$ gives the magnitude, directly for positive result, and in 2's complement form for negative result.

Try for various combinations of A and B. Verify the results.

(iii) Incrementing a number by one

Refer to line 0 of Table. Apply 0000 at the selection-lines and 0 at C_n . For various inputs at A observe the output at F. Verify the result.

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(iv) *Decrementing a number by one*

Refer to line 15 of Table. Apply 1111 at the selection-lines and 1 at Cn. For various inputs at A, observe the output at F. verify the result.

(v) *Other operations*

Verify all the remaining operations given in Table for various combinations of A and B. Verify the results.

Truth Table:

Observations:

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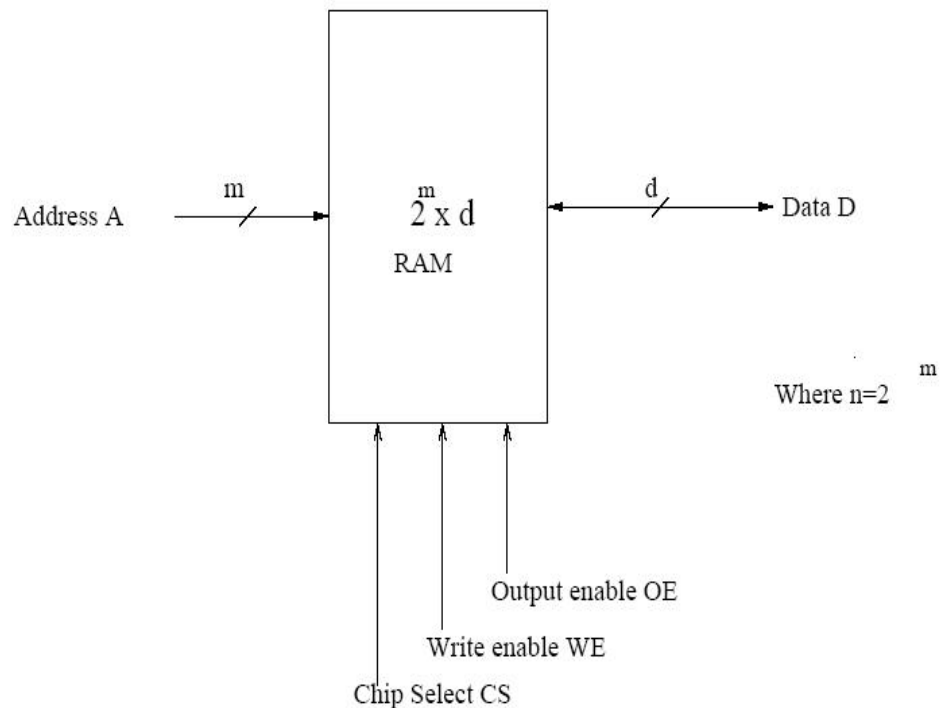
Experiment No. : 7

Title : Implement read write operations using RAM ICs.

Equipments / Components Required:

1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components
 - a. IC- 74189. (Other RAM IC can be used)
 - b. Resistors.
 - c. LEDs.

Theory: A memory with a capacity of **n** bit words is called an **$n \times d$** bit memory. That means **n** words of **d** bits each can be stored in this. A RAM IC with control lines is shown in **Fig.1**.



A RAM IC Showing its major external connections

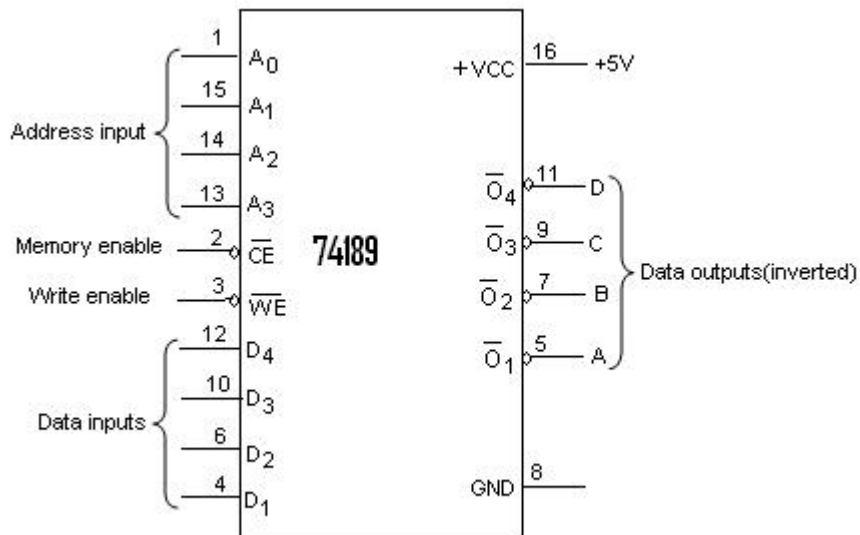
Fig.1

Here, $n=2^m$

Experimental Procedure:

- 1.Low levels refer to 0 volts and high levels refer to +5volts.
- 2.Connect the supply and Ground properly.
3. Enable the chip.
4. Write data in some address location.
5. After writing the data read the data from those address location.

Pin out diagram of a RAM chip(IC 74189):



Function Table for RAM IC 74189:

Inputs ME'	WE'	Operation	State of outputs
L	L	Write	Complements of data inputs
L	H	Read	Complements of selected word
H	L	Inhibit entry	Undetermined
H	H	Hold	High

Observations:

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Experiment No. : 8 (a) & (b).

Title : Cascade two RAM ICs for vertical and horizontal expansion.

Equipments / Components Required:

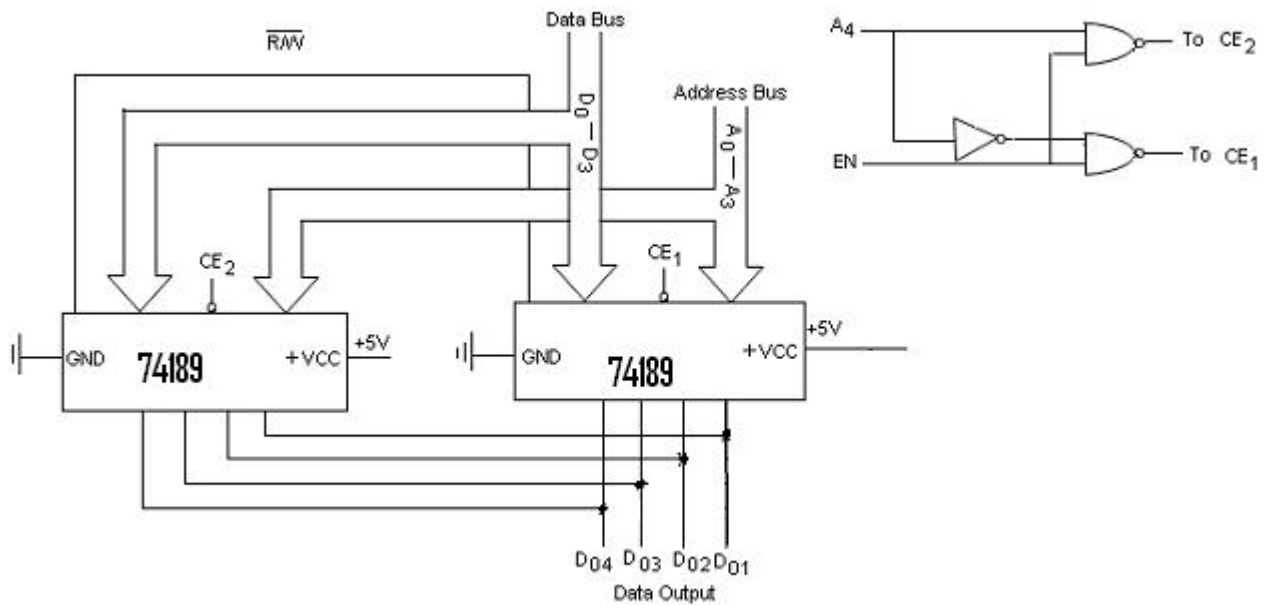
1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components
 - a. IC- 74189.(Other RAM ICs can be used)
 - b. Resistors.
 - c. LEDs.

Theory: Vertical expansion means to increase the number of locations and horizontal expansion denotes to increase the word size. A general approach is to construct a $p \times q$ array of the $M_{n,d}$ ICs where $p = \lceil n' / n \rceil$ and $q = \lceil d' / d \rceil$ and $\lceil x \rceil$ denotes the smallest integer greater than or equal to x . In this IC array each row stores n words (except possibly the last row), while each column stores a fixed set of d bits from every word (except possibly the last column). When $n' > n$, additional circuitry is usually required.

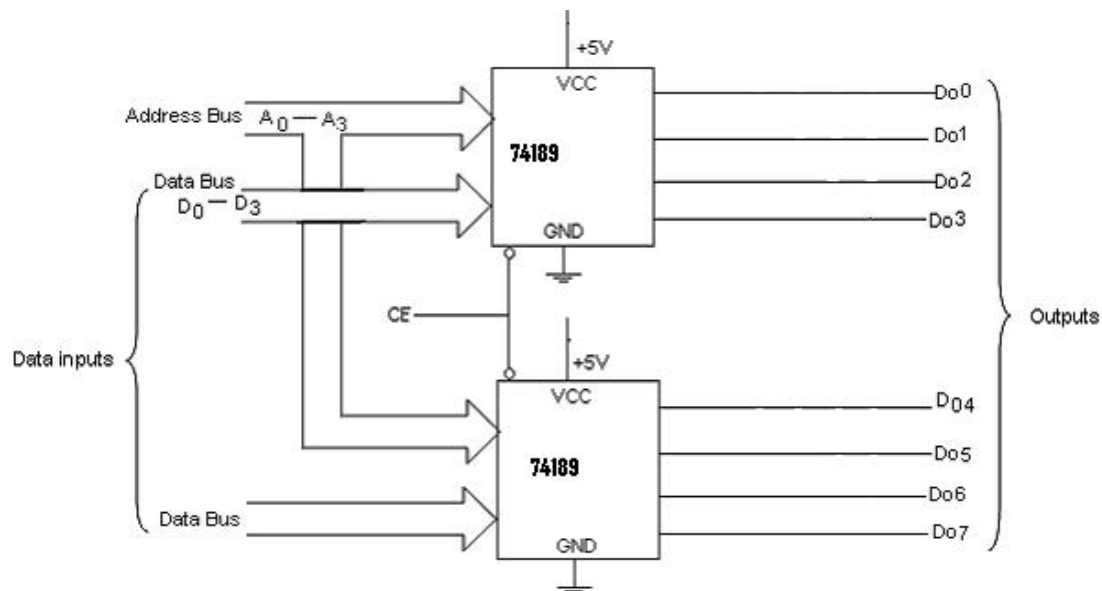
Now suppose we want to increase the number of stored words by a factor p , [i.e. $n' / n = p$] we need p RAM ICs. The number of addresses has been increased by p times hence y lines are added to the address bus (where $p = 2^y$). A one out of p address decoder must be introduced to decode the extra address bits. The original m address lines are connected to the m -bit address bus of every RAM IC; the new lines are main inputs to the decoder. Each decoder output line is connected to the CS inputs of the RAMs in the same row, ensuring that the row has a unique address. When $d' > d$, each RAM IC contains a d' -bit slice of every stored word. If $q = \lceil d' / d \rceil$ then q number ICs are needed to increase the word size q times. All the addresses and control lines are connected in exactly the same way to each IC. Their d bit buses are concatenated to form d' -bit bus. (viz d' may be $4d$, $8d$ etc.)

Circuit Diagram:

i) Vertical Expansion: (Example is given using IC 74189)



ii) Horizontal Expansion: (Example is given using IC 74189)



Observations:

Appendix

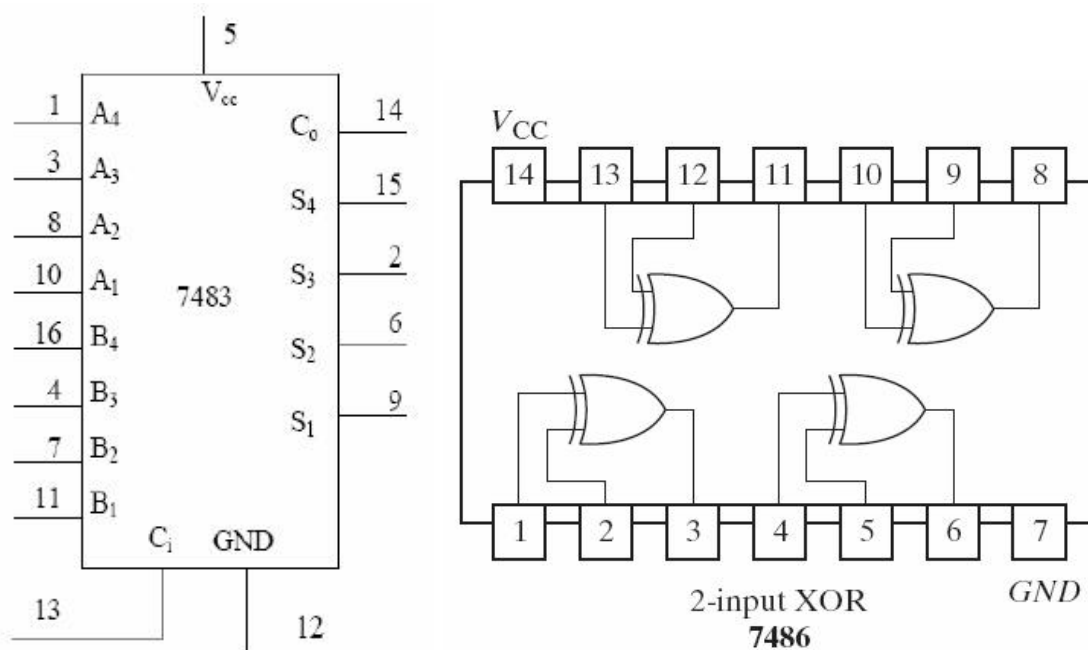
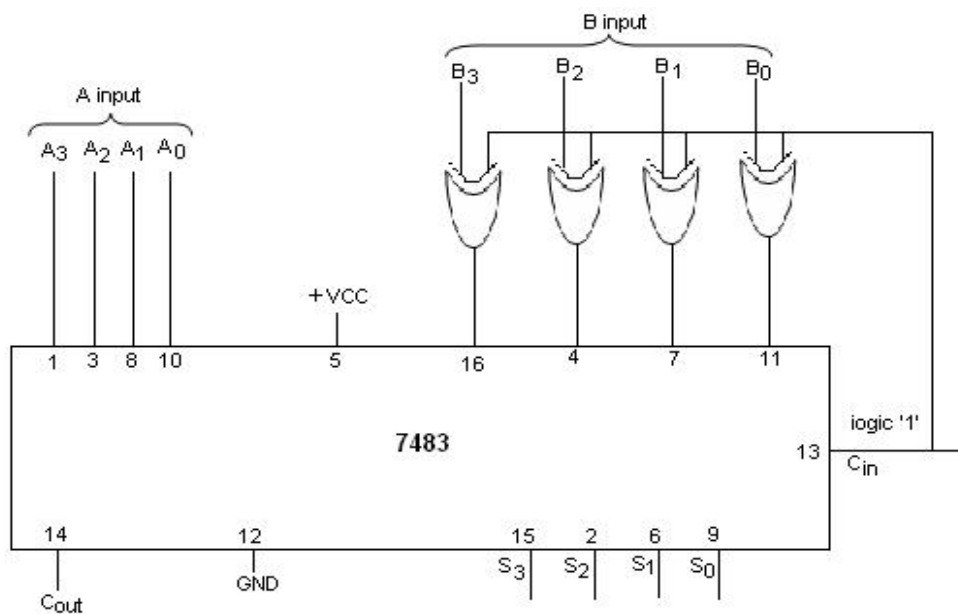


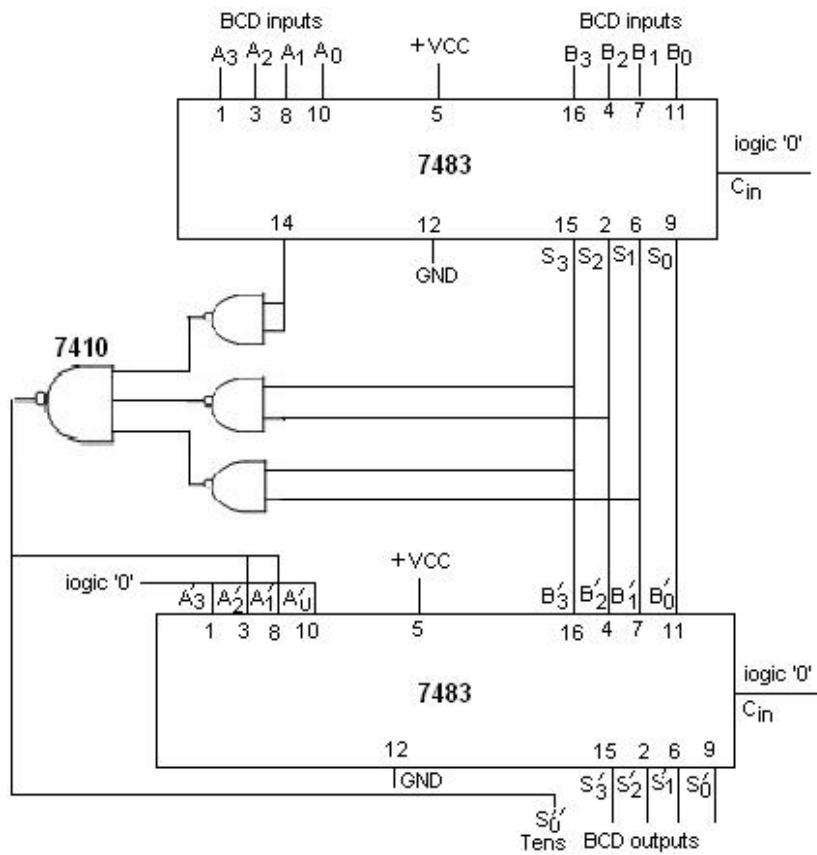
Figure 1: IC type 7483 4-bit adder.



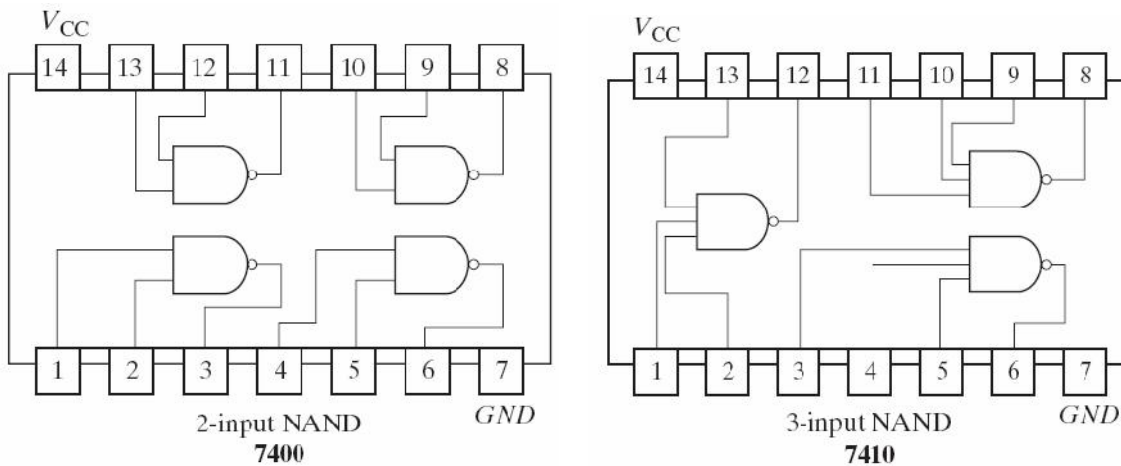
Adder- Subtractor Composite unit

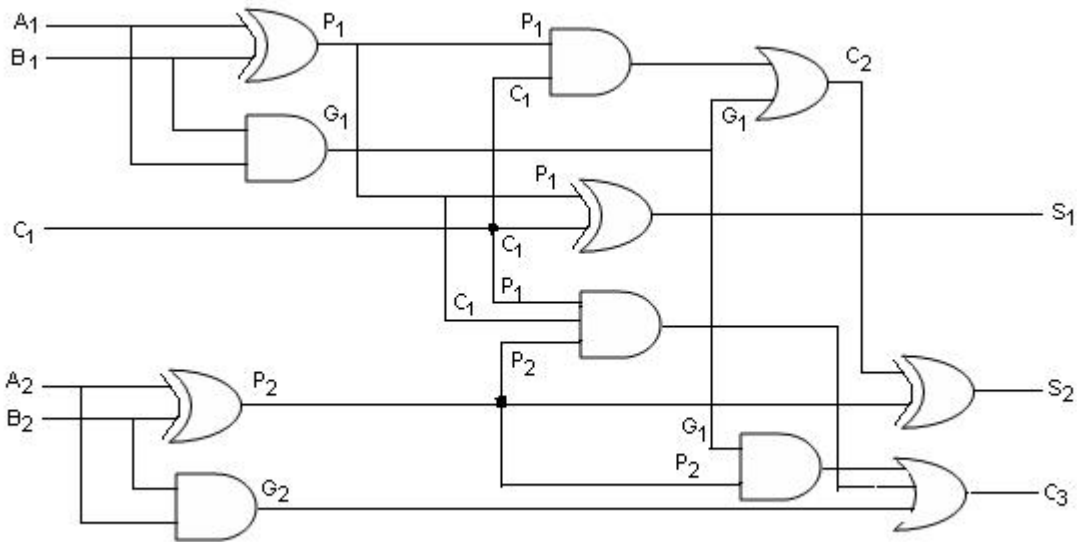
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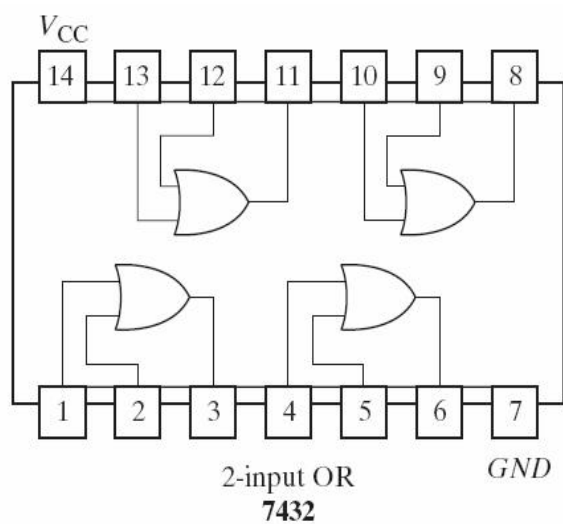
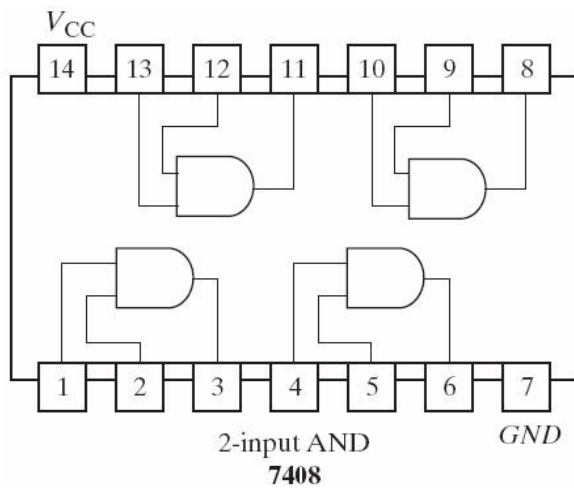


BCD adder.





2- bit carry look ahead adder.



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Experiment No. : 7

Title : Use a Multiplexer unit to design a composite ALU.

Equipments Required:

1. Bread Board.
2. +5V power supply
3. Multimeter.
4. Connecting wires.
5. Components
 - a. IC 74151.(Mention the other ICs also)
 - b. Resistors.
 - c. LEDs.

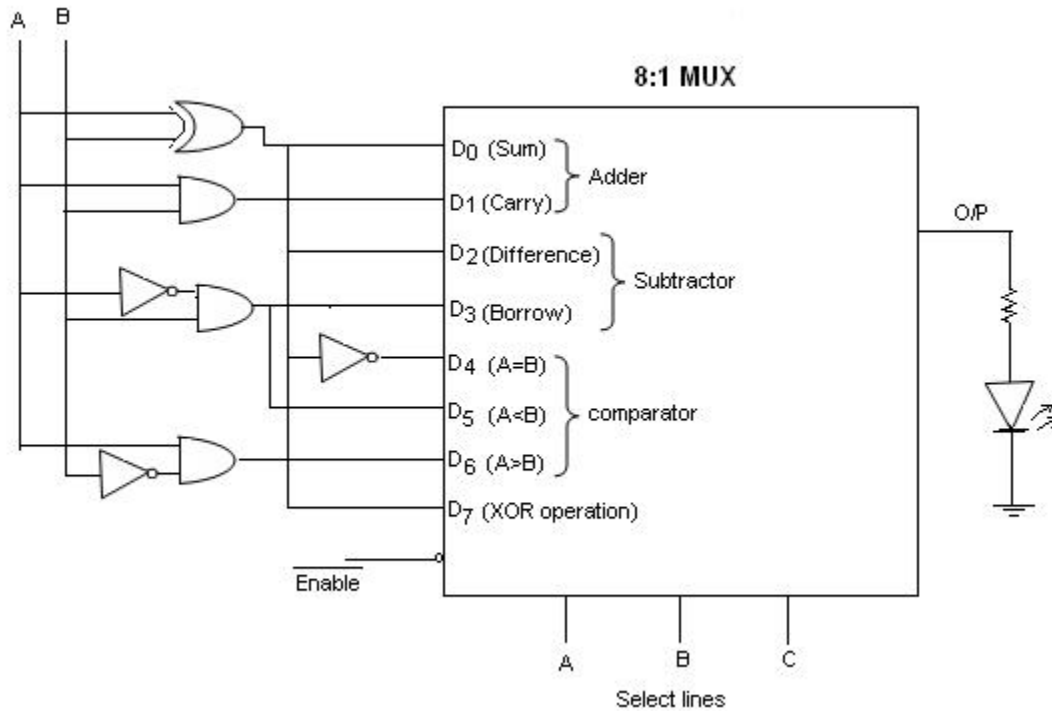
Experimental Procedure:

- 1.Low levels refer to 0 volts and high levels refer to +5volts.
- 2.Implement the circuit as per design.
- 3.Verify the truth table.

Circuit Diagram:

Implement the following function Sum, Carry, Borrow, Difference, Comparator, NAND.
Do the circuit diagram as per your design.

Observations:



Multiplexer unit to design a composite ALU.

